

Seismic imaging of coal seams and structural features: A prerequisite for CO₂ sequestration studies in the Raniganj Basin, India

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ABSTRACT

This study presents an integrated seismic and well-based characterization of the eastern Raniganj Basin, India, aimed at imaging coal seams and delineating subsurface structural features, which are essential for evaluating geological CO₂ sequestration potential. Multiple geophysical datasets, including legacy 2D seismic, high resolution 3D seismic, and 2D multicomponent (2D-3C) seismic data, were analyzed and calibrated using lithological information from drilled wells. The 3D seismic data from the northeastern part of the basin image shallow coal horizons, characterized by laterally continuous and predominantly flat lying reflectors, indicating a structurally less disturbed subsurface. In contrast, seismic interpretation in the southeastern part of the basin, based on conventional 2D and 2D-3C datasets, reveals multiple coal seams, stratigraphic boundaries, and subtle structural features, such as gentle monoclines and minor normal faults. Based on the integrated seismic analysis, the southeastern Raniganj Basin exhibits geological characteristics that are favourable for further evaluation of CO₂ sequestration potential compared to the northeastern sector. However, the presence of interpreted fault systems highlights the need for detailed investigations including high-resolution seismic imaging and fault seal analysis, to assess structural continuity and long-term containment integrity. The study establishes a preliminary understanding of the geological framework of the eastern Raniganj Basin that can be further strengthened by integrating high-resolution seismic data with other datasets to quantitatively evaluate storage capacity and long-term containment integrity.

Keywords: CO₂ sequestration, Coal seams, Seismic imaging, Subsurface structure, Raniganj Basin (India)

INTRODUCTION

Global industrialization and rapid urban expansion have led to a sustained increase in energy demand, which continues to be met predominantly by fossil fuels. While these energy resources have supported economic development, their extensive utilization has resulted in a significant rise in atmospheric CO₂ concentrations, contributing to climate change (Metz et al., 2005; Vedanti et al., 2025). This growing environmental concern, combined with the need for reliable energy, has intensified interest in cleaner and more sustainable energy solutions. In this context, geological carbon sequestration has emerged as a promising mitigation strategy (Pandey et al., 2016; Vedanti, 2026). Among various storage options, coal formations offer the additional advantage of enhanced coalbed methane recovery (ECBM), enabling simultaneous greenhouse gas mitigation and energy production (Bachu, 2008; Gale and Freund, 2001).

The Raniganj Basin (Figure 1), a sub-basin of the Damodar Valley, is of significant geological and historical importance as the birthplace of India's coal industry (Gee, 1932; Dutta, 2002). Geological investigations in the basin date back over two centuries, with formal coal mining initiated in 1774 (Varma et al., 2015). Owing to its extensive coal-bearing successions and long history of resource exploitation, the basin has attracted sustained interest in coal-bed methane and other unconventional resources. The presence of coal-based thermal power plants and ongoing coalbed methane

production by commercial operators such as Great Eastern Energy Corporation Limited (GEECL) and Essar Oil and Gas Exploration and Production Limited (EOGEL), has further motivated interest in evaluating the basin for CO₂ storage (Vedanti et al., 2020). However, safe and effective carbon sequestration in coal formations requires a detailed understanding of the subsurface geological conditions, including coal seam continuity, fault systems, and structural configuration.

Previous studies have established the regional stratigraphic and structural framework of the Raniganj Basin (Gee, 1932; Dutta, 2002; Mendhe et al., 2018; Satyanarayana et al., 2020; Satyavani et al., 2021). Multiple studies have documented the presence of several coal horizons with variable depth and thickness within the Raniganj and Barakar Formations (Ghosh, 2002; Chatterjee and Pal, 2010; Mishra et al., 2016). For instance, Mohanty et al. (2018) identified a total of nine coal horizons (R-I to R-IX) in the Raniganj Formation and seven coal horizons (I to VII) in the Barakar Formation, whereas Chattaraj et al. (2019) reported ten coal horizons (R-I to R-X) within the Raniganj Formation. These variations reflect the complex depositional and tectonic evolution of the basin.

Structurally, the basin is characterized by numerous faults, sub-basins, basement highs, and intrusive bodies (Gee, 1932; Ghosh, 2019). Intra-basinal faulting has resulted in the development of multiple structural compartments, with

spatial variability in fault trends. Earlier studies have also highlighted significant heterogeneity in subsurface structural configurations, including the presence of distinct sub-basins

in the western-central and southeastern regions (Verma et al., 1980; Ghosh et al., 1993; Kumar, 2006; Ghosh, 2019).

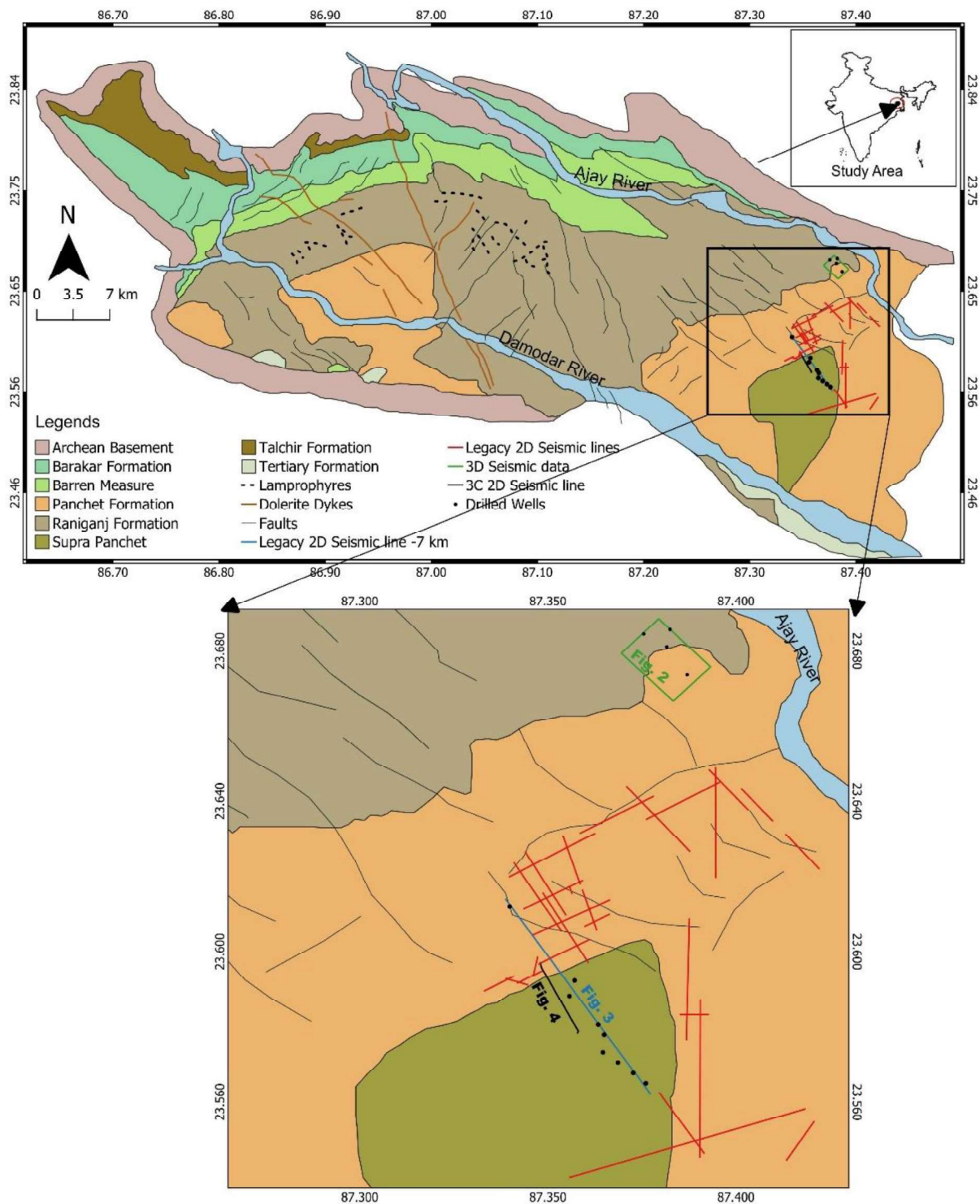


Figure 1. Geological map of the Raniganj Basin, India (modified after Murthy et al., 2010; Mohanty et al., 2018; Bandyopadhyay et al., 2020). The map highlights the locations of 2D seismic survey lines and 3D seismic data coverage

Despite these extensive investigations, subsurface information in the eastern Raniganj Basin remains limited. Uncertainties persist in imaging thin coal seams and identifying subtle fault systems, primarily due to sparse well control and limited seismic coverage. The present study addresses these gaps through a multi-dataset seismic analysis using 2D, 3D, and multicomponent (2D-3C) seismic data. The objectives are to delineate coal seams, identify structural discontinuities, and refine the regional subsurface structural framework. The results provide an improved geological understanding of coal seam distribution and structural features, forming a basis for subsequent, more detailed evaluations of CO₂ sequestration potential.

GEOLOGY OF THE RANIGANJ BASIN

Raniganj Basin is the easternmost intra-cratonic rift basin of the Damodar Valley, covering an area of approximately 1900 km² (Gee, 1932; Mendhe et al., 2018; Ghosh, 2019). The basin exhibits a semi-elliptical, elongated geometry, bounded by the Ajay River to the north and the Damodar River to the south. It is developed over a Precambrian crystalline basement composed predominantly of quartz, feldspar, and biotite (Ghosh, 2002). The sedimentary succession rests unconformably on this basement and comprises the Talchir, Barakar, Barren Measures, and Raniganj Formations, collectively representing the Lower Gondwana (Permian) sequence. These are overlain by Upper Gondwana formations (Panchet to Supra-Panchet, Triassic-Lower Cretaceous) (Fox, 1931; Mishra et al., 2016).

The Talchir Formation forms the basal unit and consists mainly of glacio-lacustrine and fluvio-glacial deposits. It is overlain by the Barakar Formation, which is characterized by sandstones, conglomerates, carbonaceous shales, and multiple coal seams, deposited in braided to fluvio-lacustrine alluvial environments (Ghosh, 2002; Mishra et al., 2016). The overlying Barren Measures Formation is a non-coal-bearing unit comprising organic-rich shales (Varma et al., 2015; Mendhe et al., 2018). The Raniganj Formation conformably overlies the Barren Measures and comprises fine-grained, micaceous sandstones, shales, and several well-developed coal seams. Structurally, the Raniganj Basin is faulted down towards the south and west, with the southern boundary defined by a series of echelon faults. The dip of the strata generally ranges between 5° and 10° (Varma et al., 2015). The geological map of the study area is shown in Figure 1.

DATA AND METHODOLOGY

This study utilizes multiple seismic and well datasets to delineate coal seams and subsurface structural features in the

eastern part of the Raniganj Basin (Figure 1). The seismic database comprises (i) Legacy 2D seismic data consisting of sixteen profiles acquired in 2007 from the south-eastern part of the basin. The survey employed an impulsive explosive source of 2.5 kg, with a receiver spacing of 10 m and a sampling interval of 0.5 ms, (ii) 3D seismic data covering an area of 2.4 km² in the north-eastern part of the basin, acquired in 2018. The survey parameters include a shot spacing of 10 m, a receiver spacing of 5 m, a sampling interval of 0.5 ms, and a frequency bandwidth of 10-150 Hz, (iii) 2D multicomponent (2D-3C) seismic data, acquired in 2022 along a 2.38 km long profile in collaboration with Essar Oil and Gas Exploration and Production Limited (EOGEP). This represents the first multicomponent seismic acquisition conducted in Indian coalfields. The acquisition employed a sampling interval of 1 ms, receiver spacing of 20 m, and shot spacing of 10 m, enabling the recording of both compressional (P-wave) and shear (S-wave) responses. (iv) Litholog data from eleven drilled wells were employed for calibration and interpretation. In addition, a density log obtained from one well was also used to support seismic interpretation.

All the seismic datasets were processed and interpreted using a standard workflow designed to enhance data quality and ensure reliable structural and stratigraphic imaging. Pre-processing steps included geometry definition, amplitude recovery, and spherical divergence correction to compensate for energy decay with depth. Predictive deconvolution and band-pass filtering were applied to suppress noise and improve temporal resolution. Velocity analysis was performed using semblance to develop stacking velocity models. These models were subsequently used for normal moveout correction and common midpoint stacking. Post-stack time migration was performed to improve reflector positioning and enhance structural continuity. The processed seismic data were then converted from time to depth using interval velocities derived from stacking analyses and calibrated with available well information. The interpretation of seismic and well data yielded a refined structural and stratigraphic framework for the eastern Raniganj Basin. This framework provides an improved basis for understanding coal seam distribution and structural configurations, supporting future and more detailed evaluations of CO₂ sequestration studies.

RESULTS

Seismic imaging of the northeastern Raniganj Basin

Two mutually perpendicular time-migrated seismic sections (Figure 2) from the 3D seismic survey in the Rangamati

Block located in the northeastern Raniganj Basin, are presented to see the lateral variations in geological structures in different directions. The results display laterally continuous and predominantly flat-lying reflections with minimal structural dip. The overall reflection pattern suggests a relatively stable and weakly deformed subsurface, with no clearly resolvable major faults or large-scale structural discontinuities within the imaged interval. A prominent high-amplitude reflection is observed at a two-way travel time of approximately 0.2 s. This reflector is characterized by a strong acoustic impedance contrast interpreted as a coal seam. Correlation with well-log and available core data confirms that this reflector corresponds to a coal seam approximately 6 m thick, occurring at a depth of ~300 m (Figure 2). Lithological interpretation based on well data indicates that the Raniganj Formation begins at a depth of ~60 m and has an overall thickness of approximately 256 m in the study area. This interpretation is consistent with previous studies, including Shaik et al. (2025), which reported multiple coal seams in the Rangamati Block.

Seismic imaging of the southeastern Raniganj Basin

Seismic results from the southeastern part of the Raniganj Basin are presented using both conventional 2D seismic and multicomponent (2D-3C) seismic datasets to delineate coal

seams and subsurface structures. A depth-scaled seismic section along a ~7 km long 2D seismic line oriented in a NW-SE direction (location shown in Figure 1) is presented in Figure 3. The shallow portion of the section, up to approximately 400 m, was muted due to high noise levels. Below this depth, multiple subsurface reflectors are observed, with stratigraphic layering becoming more distinct between depths of ~600 and ~1200 m. Reflections within this interval exhibit improved continuity and amplitude coherence. High-amplitude, high-contrast reflectors are interpreted as coal seams and are validated through correlation with well-log data and stratigraphic markers (Figure 3). Based on well control, the Raniganj Formation is identified at depths of approximately 600-700 m, with a thickness ranging from ~500 to 700 m along the profile. The Barren Measures Formation occurs at depths of ~1100-1400 m; however, its base could not be mapped due to limited seismic penetration. Structurally, the reflectors define a broad southeast-dipping monocline. Individual coal seams (Seams 1, 2, and 3) are separated by non-coaly interbeds of sandstone or shale. The reflectors deepen towards the central part of the seismic section (offsets ~1-3.5 km) and become shallower towards the flanks, suggesting a broad antiformal-synform geometry with superimposed minor undulations. Several small-scale normal faults are interpreted, locally affecting the reflector continuity.

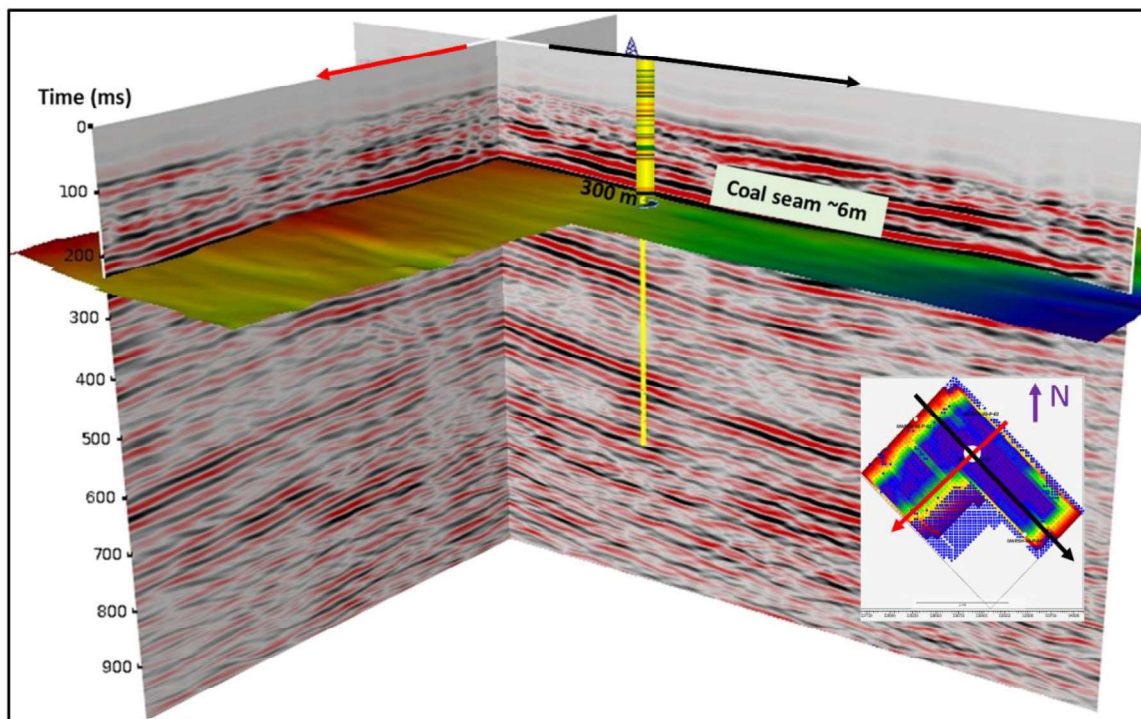


Figure 2. Two mutually perpendicular time-migrated seismic sections from the 3D seismic survey. High-amplitude reflections near 0.2 s indicate strong acoustic impedance contrast, interpreted as coal seams and confirmed by the litholog of the drilled well.

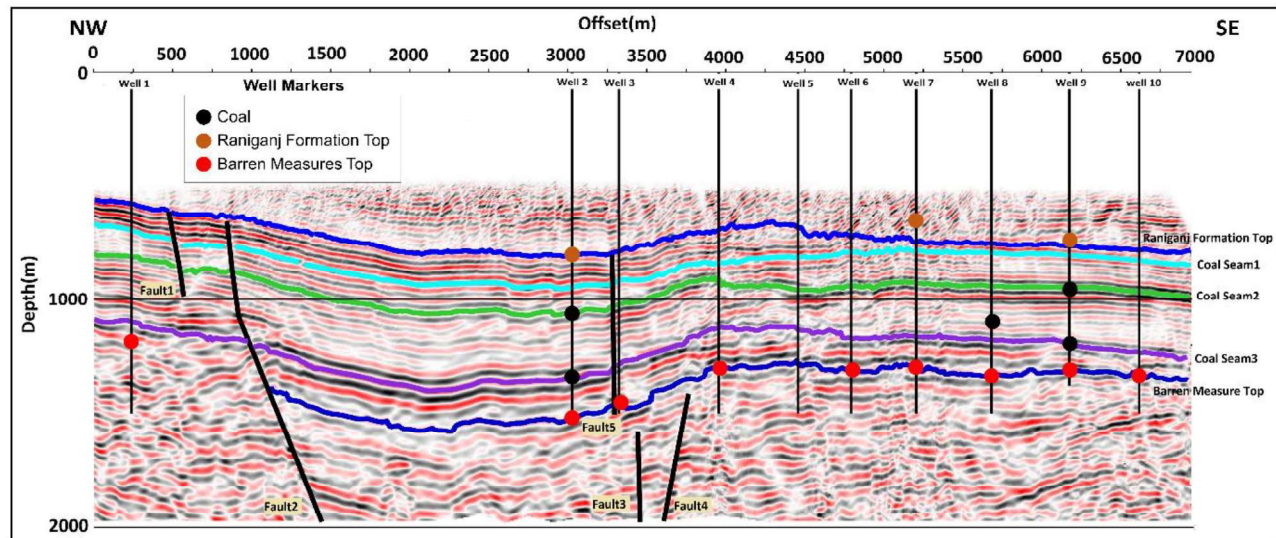


Figure 3. A depth-scaled seismic section showing the presence of coal seams, minor faulting, and broadly folded layers

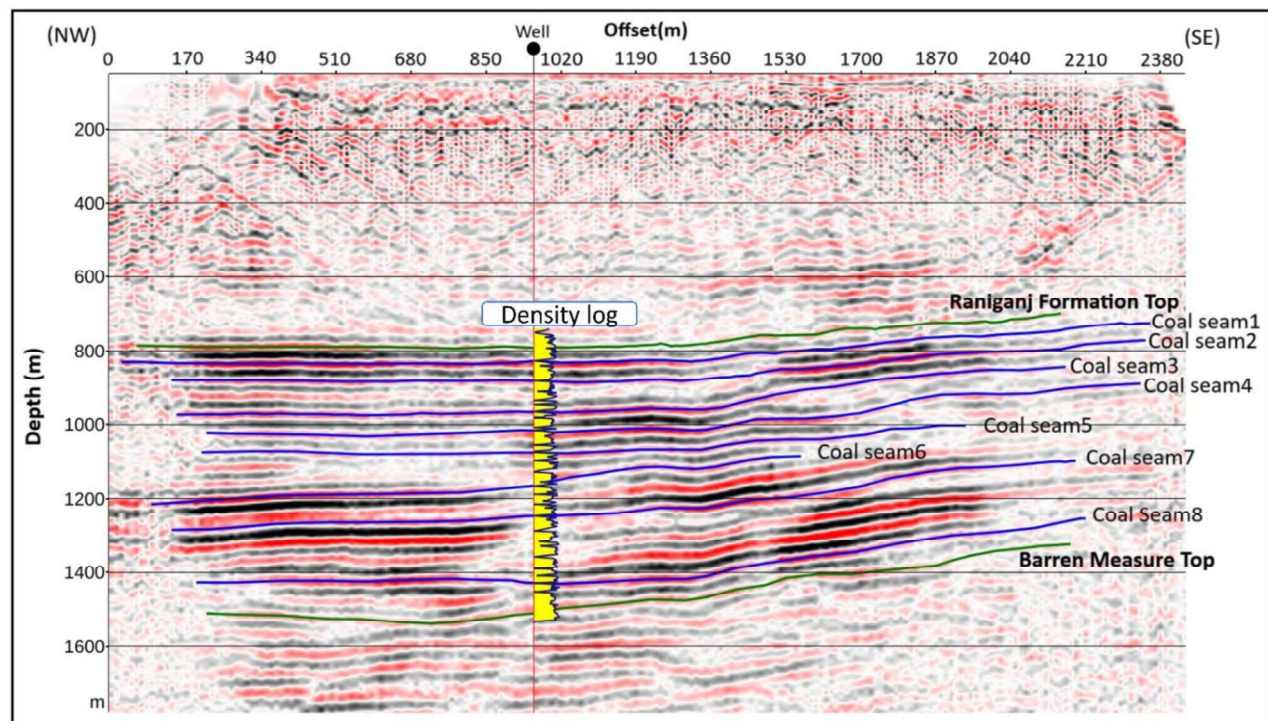


Figure 4. Depth-scaled seismic section along a 2D profile in the Raniganj (East) block showcasing distinct reflectors. High-amplitude contrast suggests the presence of coal layers that are validated with the density log data of a drilled well.

A depth-scaled seismic section derived from the 2D-3C seismic data along a ~2.4 km long NW-SE-oriented profile (location shown in Figure 1) in the southeastern part of the basin is presented in Figure 4. This section exhibits distinct amplitude variations associated with lithological contrasts, particularly within coal-bearing intervals. We used density log data from a borehole located along the seismic profile to

identify coal seams as it is characterized by low-density signatures (Figure 4). Correlation between seismic and density -log data confirms the presence of multiple coal seams. The well data indicate six major coal seams, each ~3 m thick, and several thin coal seams (~1-2 m). However, only eight coal seams could be confidently interpreted on the seismic section due to resolution limitations. The Raniganj

Formation occurs at a depth of ~700 m in the southeastern part of the profile and deepens to ~800 m towards the northwest, indicating a gentle northwestward dip. The formation exhibits a thickness of ~600-700 m along the profile. The Barren Measures Formation is encountered at depths of ~1350 to 1500 m and displays a structural configuration similar to the overlying Raniganj Formation.

DISCUSSION

Subsurface structures in the eastern Raniganj Basin

The combined interpretation of 2D, 3D, and multicomponent seismic datasets reveal significant spatial variability in subsurface structural architecture across the eastern Raniganj Basin. The northeastern sector, particularly the Rangamati Block, is characterized by shallow coal seams associated with laterally continuous, sub-horizontal reflectors and minimal structural deformation. Although previous studies (Chatterjee and Pal, 2010; Sen et al., 2016; Ahmed et al., 2025) have reported shallow coal occurrence in this region, constraints on the detailed structural framework have remained limited. The present results indicate that coal horizons within the Rangamati Block, are laterally continuous and exhibit negligible structural disruption over the extent of the 3D seismic survey (~2.4 km²). However, this interpretation is spatially restricted to the surveyed area, and structural variations beyond this extent cannot be ruled out.

In contrast, the southeastern part of the basin exhibits deeper coal horizons and a comparatively more complex structural configuration, as evidenced by gentle monoclines, broad antiformal-synform geometries, and small-scale normal faults. Sen et al. (2016) have documented relatively deeper coal seams in the central and southern sectors of the eastern Raniganj Basin, and highlighted significant depth variability based on litholog data from multiple wells. Earlier geological interpretations (Verma et al., 1980; Ghosh et al., 1993; Kumar, 2006), proposed the existence of a sub-basin in this region, commonly referred to as the Durgapur sub-basin. The observed depth variations and structural geometries in the present study are consistent with earlier interpretations and suggest the presence of a basin-like structural configuration or a sub-basin in the study area. Recent findings by Bandyopadhyay et al. (2025) further support the presence of gentle deformation and faulting within coal-bearing sequences.

Overall, the results indicate the coal seams in the northeastern sector are shallow, laterally continuous, and structurally less disturbed, whereas those in the southeastern sector occur at greater depths and exhibit increased structural variability.

These observations are consistent with earlier studies (Ghosh, 2002; Sen et al., 2016; Ghosh, 2019), which report systematic deepening of coal seams away from basin margins and towards structurally depressed regions. For instance, Sen et al. (2016) reported coal seams at relatively shallower depths (~450 m) near the basin margins, compared to deeper occurrences (~650 m) towards the basin interior. Similarly, Ghosh (2002, 2019) demonstrated that coal seams tend to be thicker and deeper in the central part of the basin, while thinning or terminating toward the margins.

Implications for CO₂ sequestration potential

The observed variations in burial depths and structural configurations across the eastern Raniganj Basin have important implications for evaluating CO₂ sequestration potential. In the northeastern region, coal seams occur at relatively shallow depths, which are generally considered less favourable for long-term CO₂ storage due to insufficient pressure-temperature conditions, and the possibility of leakage to surface or interaction with shallow groundwater systems. Previous CCS studies suggest that effective CO₂ sequestration in coal seams typically requires burial depths exceeding 800 m to ensure supercritical CO₂ conditions and enhanced storage efficiency (White et al., 2005; Bachu, 2008).

In contrast, the southeastern sector is characterized by deeper coal-bearing horizons and thicker stratigraphic successions, which are more consistent with conditions generally considered suitable for CO₂ storage. However, the presence of interpreted fault systems and structural heterogeneities introduces uncertainty regarding subsurface continuity and potential fluid migration pathways. These factors necessitate careful evaluation of fault seal integrity, caprock effectiveness, and geomechanical stability prior to any site-specific assessment of CO₂ storage feasibility.

The results demonstrate the utility of seismic data for delineating coal seams and resolving subsurface structural features relevant to CO₂ storage feasibility. At the same time, limitations in resolving thin coal seams highlight the need for advanced approaches such as seismic inversion, full-waveform inversion, and joint geophysical-petrophysical modelling. Overall, the southeastern Raniganj Basin exhibits geological characteristics that warrant further detailed investigations for CO₂ sequestration. Future studies should integrate seismic-derived structural models with coal petrophysical properties, adsorption characteristics, and in-situ stress characterization to enable a robust assessment of storage capacity and long-term containment integrity.

CONCLUSIONS

This study presents a multi-dataset seismic analysis of coal seams and subsurface structures in the eastern Raniganj Basin, supported by well data, with implications for geological CO₂ sequestration. The key findings are as follows.

1. The combined analysis of 3D, 2D, and 2D-3C seismic data, calibrated with well logs and core information, enables reliable delineation of coal seams, formation boundaries, and structural features relevant to subsurface characterization.
2. The northeastern part of the basin is characterized by laterally continuous and weakly deformed coal-bearing sequences, whereas the southeastern part exhibits deeper coal horizons associated with gentle monoclines and minor faulting, indicating spatial variability in structural conditions.
3. Based on the observed variations in seam continuity, burial depth, and structural configuration, the southeastern Raniganj Basin exhibits geological characteristics that warrant further detailed evaluation for CO₂ sequestration. However, the presence of interpreted fault systems necessitates additional investigations, including assessments of fault-seal integrity and caprock effectiveness, before confirming its suitability for long-term CO₂ storage.

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Data availability

Data is available at CSIR-NGRI and will be shared on request.

Compliance with ethical standards

No conflict of interest, and authors adhere to copyright norms.

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